Highlights

Risk-aware trajectory planning for fixed-wing UAVs via multi-objective two-stage search and flight-envelope-constrained refinement

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- Time, exposure and fuel share fixed references across two search resolutions.
- Fine-grid replanning resolves a coarse route within a three-dimensional corridor.
- Spline geometry and assigned speeds undergo explicit flight-envelope checks.
- Final mission costs are recomputed after geometric refinement and retiming.

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Abstract

Fixed-wing mission planning in spatially varying threat environments requires decisions about where to fly and how to traverse the selected route. A collision-free path alone does not specify the exposure accumulated during flight or its compatibility with aircraft motion limits. This paper presents a risk-aware planning framework that connects mission-dependent route selection with flight-envelope-constrained trajectory generation. Hard no-fly regions are excluded from the feasible space, while a continuous radar-exposure surrogate is integrated over flight time. Time, exposure and fuel estimates are combined using shared normalization references obtained from three single-objective coarse searches. A coarse A* search establishes the route, followed by fine-grid replanning within a three-dimensional corridor under the same weighted objective. Cubic B-spline candidates are then evaluated using analytical curvature and sampled geometric checks. Forward–backward speed assignment enforces the modeled speed, turning, vertical-rate and tangential-acceleration limits. Mission metrics are recomputed along the final trajectory using the resulting travel times. The evaluation protocol combines controlled objective and stage comparisons with a 72-case preference study in a fixed five-zone environment, separating full two-stage evaluations from coarse-search sensitivity cases. The framework provides a traceable connection between planning preferences, geometric refinement and final mission evaluation within a reduced fixed-wing flight-envelope model.

Keywords: Fixed-wing UAV, trajectory planning, risk-aware planning, multi-objective optimization, A* search, B-spline

1. Introduction

1.1. Background and motivation

Fixed-wing unmanned aerial vehicle (UAV) mission planning involves coupled decisions about route geometry, travel time and energy expenditure. In a threat environment, these decisions also determine the duration and intensity of exposure. Two collision-free routes can therefore have markedly different mission costs: a detour may avoid a region of high threat intensity but require more time and fuel, whereas a shorter route may cross that region for a limited duration. A useful planner must distinguish absolute prohibitions from traversable regions whose desirability depends on the mission preference.

A* provides a general mechanism for searching a graph with assigned edge costs [1]. Its capacity to represent soft threats depends on the chosen graph and cost model. In a binary-occupancy, time-oriented formulation, obstacle checks establish admissibility and travel-time estimates rank the remaining routes; different threat intensities within free space do not affect that ranking unless they are included in the cost. The question addressed here is how to combine this spatial information with time and fuel preferences while retaining a clear connection to the trajectory that will ultimately be evaluated.

Three issues arise in this connection. First, exposure is accumulated over time, so a spatial threat score alone cannot determine the cost of a variable-speed trajectory. Second, increasing the resolution of a three-dimensional search expands its computational requirements, motivating local refinement of a coarse route. Third, collision-free grid edges can meet at corners that a fixed-wing aircraft cannot follow directly. Smoothing changes the geometry, and speed assignment changes the time spent along it. Consequently, the mission metrics of the final trajectory must be distinguished from the estimates used to select the discrete path.

1.2. Related work

Risk-aware and multi-objective planning incorporate more information than binary occupancy. Guo et al. [2] combine threat strength and path length in FC-RRT*, using the resulting cost and flight constraints in tree expansion and parent selection. Their formulation illustrates that traversable threats and mission-dependent trade-offs are established planning concepts.

For the present problem, the modeling emphasis is on time-integrated exposure together with flight time and fuel consumption. Because subsequent speed assignment can alter exposure even when the route is unchanged, these quantities also require a common final evaluation protocol.

Hierarchical planning addresses the cost of resolving large spatial domains. Liu et al. [3] couple a multigranularity semantic-risk representation with coarse corridor guidance and fine-level A* search. Their urban framework also supports replanning after airspace changes; its voxel-center output requires subsequent smoothing and platform-specific dynamic verification. This work establishes a close precedent for risk-aware corridor refinement. The present study concentrates on fixed-wing trajectory generation after route selection, carrying shared time–exposure–fuel references through both search resolutions and evaluating the refined trajectory under its assigned speed profile.

Fixed-wing trajectory methods explicitly address the relation between geometry and motion. Li et al. [4] decompose constrained trajectory optimization into path and speed subproblems and use nonlinearity transfer and convexification to obtain convergent solution procedures. Zhang et al. [5] use an improved A* path to initialize states and controls for subsequent pseudospectral trajectory optimization. These studies show that path–speed decomposition and search-to-trajectory integration already have substantive precedents. Here, a reduced flight-envelope formulation is used to connect preference-based graph search with spline refinement and speed assignment, rather than to solve a full aircraft optimal-control problem. Parametric spline fitting itself follows established methods [6]; the application-specific issue is how the fitted curve is constrained, selected and reevaluated.

1.3. Research focus and contributions

The problem addressed in this paper is the consistency of the transition from mission-dependent route selection to a constrained fixed-wing trajectory. This consistency has three concrete requirements: the two search resolutions must use the same objective references; the smoothed curve must be checked against the constraints affected by geometric modification; and final time, exposure and fuel must be calculated using the assigned motion rather than inherited from the search. The proposed framework addresses these requirements through the following contributions.

1. A mission-preference-driven time–exposure–fuel formulation is constructed

for hard–soft threat environments. Hard no-fly regions define infeasibility, while a continuous radar-exposure surrogate enters the cost through its time integral. Three single-objective coarse-search anchors provide fixed reference scales for weighted route selection and subsequent reporting.

2. A coarse-to-fine three-dimensional planning procedure applies the same weighted search objective and reference scales at both resolutions. The coarse route defines a spatial corridor within which a fresh fine-grid search resolves local geometry. This procedure makes the role of spatial refinement explicit without interpreting interpolation as replanning.
3. A constrained path-to-trajectory procedure connects cubic B-spline refinement, analytical curvature evaluation, sampled feasibility checks and forward–backward speed assignment. Final mission costs are reintegrated along the refined trajectory, exposing the changes introduced by geometric modification and retiming instead of treating discrete-path statistics as executable-trajectory performance.

The numerical study is organized around controlled comparisons in one prescribed environment. Full two-stage cases examine the objective choice and the successive planning stages, while a separate 72-case coarse-search protocol characterizes preference sensitivity and supplies representative trajectories. This design separates evidence about the complete planning chain from evidence about weight-dependent behavior. Feasibility throughout this paper refers to the specified reduced flight envelope and numerical checks; it does not imply a full flight-dynamics or tracking validation.

Section 2 defines the environment, constraints and mission objective. Section 3 presents the planning and evaluation procedure. Section 4 specifies the controlled comparisons and preference analysis, and Section 5 concludes the paper.

2. Problem formulation

2.1. Three-dimensional mission and threat environment

The mission is defined in a three-dimensional Cartesian workspace $\mathcal{W} \subset \mathbb{R}^3$. The UAV position is $\mathbf{p} = [x, y, z]^\mathsf{T}$, where x and y are horizontal coordinates and z is altitude. The prescribed mission endpoints are $\mathbf{p}_s$ and $\mathbf{p}_g$. A discrete search path is written as

$$\Gamma_d = (\mathbf{p}_0, \ldots, \mathbf{p}_N), \qquad \mathbf{p}_0 = \mathbf{p}_s, \quad \mathbf{p}_N = \mathbf{p}_g, \tag{1}$$

whereas the final time-parameterized trajectory is $\mathbf{r}(t) \in \mathcal{W}$, $t \in [0, T]$. These representations are distinguished because collision-free graph connectivity does not establish continuous-trajectory feasibility. The search, geometric refinement and speed-assignment procedures are described in Section 3.

Hard exclusion regions and traversable soft-threat regions are modeled separately. Each hard zone is a vertical cylinder,

$$\Omega_{F,j} = \left\{ \mathbf{p} \in \mathcal{W} : (x - x_{F,j})^2 + (y - y_{F,j})^2 \le R_{F,j}^2, \quad 0 \le z \le H_{F,j} \right\}, \tag{2}$$

where $R_{F,j}$ and $H_{F,j}$ are its horizontal radius and height. With $\Omega_{\text{hard}} = \bigcup_{j=1}^{N_F} \Omega_{F,j}$, the continuous collision-avoidance requirement is

$$\mathbf{r}(t) \notin \Omega_{\text{hard}}, \qquad t \in [0, T]. \tag{3}$$

For radar i at $(x_{R,i}, y_{R,i}, 0)$, define the slant range

$$d_i(\mathbf{p}) = \sqrt{(x - x_{R,i})^2 + (y - y_{R,i})^2 + z^2}. \tag{4}$$

Its coverage is hemispherical within the nonnegative-altitude workspace. Let R_i be its outer radius and $R_{\text{in},i} = \eta R_i$, with $0 < \eta < 1$. For the unit radar scaling factors used in the experiments, the local exposure field is

$$P_{d,i}(\mathbf{p}) = \begin{cases} 1, & d_i(\mathbf{p}) \le R_{\text{in},i}, \\ \left(\dfrac{R_i - d_i(\mathbf{p})}{R_i - R_{\text{in},i}} \right)^{\nu}, & R_{\text{in},i} < d_i(\mathbf{p}) < R_i, \\ 0, & d_i(\mathbf{p}) \ge R_i, \end{cases} \tag{5}$$

where $\nu > 0$ controls the decay rate. The experiments use $\eta = 0.3$ and $\nu = 4$. Overlapping fields are combined by

$$P_d(\mathbf{p}) = \max_{1 \le i \le N_R} P_{d,i}(\mathbf{p}), \qquad 0 \le P_d(\mathbf{p}) \le 1. \tag{6}$$

This is a prescribed dominant-exposure surrogate, rather than a statistical fusion formula for the probability of detection by at least one radar. It does not model radar scan timing, track accumulation or measurement dependence. Hard zones determine admissibility, while the soft field contributes to the accumulated exposure objective in Section 2.3.

2.2. Fixed-wing flight constraints

The model assumes still air and constant aircraft mass. Hence the scalar path speed $v(t) = \|\dot{\mathbf{r}}(t)\|$ is identified with airspeed. Feasibility refers to the specified geometric and reduced flight-envelope constraints; it does not imply validation of a six-degree-of-freedom aircraft model or a closed-loop tracking controller.

2.2.1. Mission altitude envelope

Let $d_s(\mathbf{p})$ and $d_g(\mathbf{p})$ denote the horizontal distances to the start and goal. For transition distance D_{tr}, define

$$\alpha_e(\mathbf{p}) = \mathrm{clip}\left(\frac{d_e(\mathbf{p})}{D_{\mathrm{tr}}}, 0, 1\right), \qquad e \in \{s, g\}, \tag{7}$$

and $h_e(\mathbf{p}) = z_e + (h_{\mathrm{en}} - z_e)\alpha_e(\mathbf{p})$. Here h_{en} is the minimum en-route altitude. The local envelope is

$$h_{\min}(\mathbf{p}) = \min\{h_s(\mathbf{p}), h_g(\mathbf{p}), h_{\mathrm{en}}\}, \qquad h_{\min}(\mathbf{r}(t)) \le z(t) \le h_{\max}. \tag{8}$$

This construction retains the endpoint altitudes while imposing the en-route lower bound outside both transition regions. A sufficient compatibility condition for the ramp slope is

$$D_{\mathrm{tr}} \ge \max_{e\in\{s,g\}} \frac{\max(h_{\mathrm{en}} - z_e, 0)}{\tan\gamma_{\max}}. \tag{9}$$

This condition concerns the altitude envelope itself; it does not guarantee that a feasible route exists in the presence of obstacles. In the present mission, $z_s = z_g = 0$ is an altitude reference for airborne trajectory segments. Ground roll, take-off dynamics and landing dynamics are not modeled.

2.2.2. Airspeed constraint

The aircraft must remain above its stall-speed lower bound. The speed assignment uses a prescribed cruise ceiling v_{cr}, additionally bounded by the aircraft maximum speed $v_{\max}$:

$$v_{\mathrm{stall}} \le v(t) \le v_{\mathrm{cap}}, \qquad v_{\mathrm{cap}} = \min\{v_{\mathrm{cr}}, v_{\max}\}. \tag{10}$$

For the reported configuration, $v_{\mathrm{stall}} = 33\ \mathrm{m\,s^{-1}}$, $v_{\mathrm{cr}} = 55\ \mathrm{m\,s^{-1}}$ and $v_{\max} = 83\ \mathrm{m\,s^{-1}}$. The airborne initial condition is $v(0) = v_{\mathrm{stall}}$. No terminal-speed equality is prescribed; $v(T)$ follows from the local ceilings and the forward–backward speed assignment. The current speed assignment seeks the available speed under these ceilings and does not solve a separate fuel-optimal speed-control problem.

2.2.3. Turning constraint

For a coordinated level turn, the bank-limited radius satisfies

$$R_{\min}(v) = \frac{v^2}{g \tan \phi_{\max}}. \tag{11}$$

The implementation applies this relationship as a curvature-based envelope test on the three-dimensional path. For a regular spline $\mathbf{q}(u)$, its spatial curvature is

$$\kappa(u) = \frac{\|\mathbf{q}'(u) \times \mathbf{q}''(u)\|}{\|\mathbf{q}'(u)\|^3}, \qquad v_{\text{turn}}(u) = \sqrt{\frac{g \tan \phi_{\max}}{\kappa(u)}}. \tag{12}$$

The turning ceiling is interpreted as infinite where $\kappa = 0$. The assigned speed must not exceed both v_{cap} and v_{turn}. If $v_{\text{turn}} < v_{\text{stall}}$, the geometry is rejected; the speed is not reduced below stall to make it acceptable. Using spatial curvature in this level-turn relationship is a reduced-model approximation for gently climbing trajectories, not an exact bank-angle reconstruction for general three-dimensional flight. A constant stall threshold is used; bank-dependent stall speed and full load-factor dynamics are outside the present model.

2.2.4. Climb and vertical-rate constraints

With horizontal arc length s_h, the flight-path angle and vertical rate obey

$$\gamma = \arctan\left(\frac{\mathrm{d}z}{\mathrm{d}s_h}\right), \qquad |\gamma| \le \gamma_{\max}, \qquad -v_{\text{des,max}} \le \dot{z} \le v_{\text{asc,max}}. \tag{13}$$

The angle bound restricts geometry, while $\dot{z} = v \sin \gamma$ couples that geometry to speed. Both are required. The current limits are $\gamma_{\max} = 15°$, $v_{\text{asc,max}} = 12\ \mathrm{m\,s^{-1}}$ and $v_{\text{des,max}} = 15\ \mathrm{m\,s^{-1}}$.

2.2.5. Longitudinal acceleration and numerical feasibility

The tangential acceleration is bounded by

$$-a_{\text{dec,max}} \le \dot{v}(t) \le a_{\text{acc,max}}, \tag{14}$$

with $a_{\text{acc,max}} = 3\ \mathrm{m\,s^{-2}}$ and $a_{\text{dec,max}} = 4\ \mathrm{m\,s^{-2}}$ in the experiments. These bounds are imposed during speed assignment rather than added to the cost.

Search edges are screened for environmental and geometric feasibility. After spline construction, endpoint retention, terminal horizontal heading, local deviation, altitude, obstacle clearance and curvature are checked again, followed by speed-envelope validation. The terminal reference heading is inferred from the incoming discrete path, rather than supplied as an independent mission heading. The fine-search corridor restricts graph search; final spline locality is checked against the input path, and membership in the coarse corridor is diagnostic by default.

The continuous constraints above specify the intended model. Their implementation uses finite-resolution sampling and numerical tolerances: the spline geometry is probed at 4096 parameter values, endpoints are accepted within 2 m, and the terminal-heading tolerance is 12°. Therefore, acceptance denotes satisfaction of these numerical tests, not a formal continuous-domain safety certificate. Numerical settings and resolution checks must accompany the reported experiments.

2.3. Time, threat-exposure, and fuel models

The mission metrics are flight time T, accumulated exposure C, and fuel consumption F. They are evaluated on a common trajectory and with the same time increments. Let $s \in [0, L]$ denote three-dimensional arc length along the final curve and let $v(s) > 0$ be its assigned path speed. The associated time law is $\mathrm{d}t = \mathrm{d}s/v(s)$.

2.3.1. Flight time

The continuous flight-time functional is

$$T = \int_0^L \frac{1}{v(s)}\,\mathrm{d}s. \tag{15}$$

For final trajectory samples $\mathbf{q}_i$ with speeds v_i, the numerical evaluator uses the chord length $\ell_i = \|\mathbf{q}_{i+1} - \mathbf{q}_i\|$ and constant tangential acceleration on each sampled segment. Consequently,

$$\Delta t_i = \frac{2\ell_i}{v_i + v_{i+1}}, \qquad a_i = \frac{v_{i+1}^2 - v_i^2}{2\ell_i}, \qquad T_{\mathrm{num}} = \sum_i \Delta t_i, \tag{16}$$

for nonzero-length segments. This is exact for the adopted segment motion model, while the sampled representation approximates the continuous spline. It avoids replacing the final trajectory speed by one mission-wide constant.

During graph search, the speed profile is not yet available. A straight edge e of length ℓ_e and altitude change Δz_e instead uses

$$\Delta t_e^{\text{search}} = \max\left\{\frac{\ell_e}{v_{\text{cr}}}, \frac{|\Delta z_e|}{v_{z,\max}(e)}\right\}, \tag{17}$$

where $v_{z,\max}(e)$ is the ascent or descent limit according to the sign of Δz_e; the second term is zero for level flight. Here v_{cr} lies within the aircraft speed interval. This edge model is a search surrogate and does not account for acceleration across graph vertices. The final metrics are recomputed after geometric refinement and speed assignment; they need not equal the search-stage estimates.

2.3.2. Accumulated threat exposure

The exposure objective is

$$C = \int_0^T P_d(\mathbf{r}(t))\,\mathrm{d}t = \int_0^L \frac{P_d(\mathbf{q}(s))}{v(s)}\,\mathrm{d}s. \tag{18}$$

Because P_d is dimensionless and bounded by unity, C has units of seconds and satisfies $0 \le C \le T$. It measures time-weighted exposure, not mission detection probability, interception probability or expected loss. The same geometric passage can incur different exposure when the time spent within the field changes. Thus a length integral of P_d alone would not represent the objective evaluated here.

2.3.3. Fuel consumption

Let $v_h = \sqrt{\dot{x}^2 + \dot{y}^2}$ and $v_z = \dot{z}$ be the horizontal and vertical velocity components. The implemented fuel model uses a quasi-steady parabolic-drag power surrogate. Air density is approximated by

$$\rho(z) = \rho_0 \exp(-z/H_\rho), \qquad \rho_0 = 1.225\ \text{kg}\,\text{m}^{-3}, \quad H_\rho = 8500\ \text{m}. \tag{19}$$

This is an exponential atmosphere approximation. It is not the full piecewise International Standard Atmosphere model. The level-flight mechanical power surrogate is

$$P_{\text{lev}}(v_h, z) = \frac{1}{2}\rho(z) C_{D0} S v_h^3 + \frac{2(mg)^2}{\rho(z) v_h S \pi\,\text{AR}\,e_O}, \tag{20}$$

where S is wing area, C_{D0} is the zero-lift drag coefficient, AR is aspect ratio, and e_O is the Oswald efficiency factor. The first and second terms represent parasitic and induced power, respectively. The model uses horizontal speed in these terms; it does not substitute path speed v for v_h in climbing segments.

For the current configuration, descent energy recovery is zero. With climb efficiency η_c and fuel-to-propulsive efficiency η_p, the fuel-equivalent power is

$$P_f(v_h, v_z, z) = \frac{P_{\text{lev}}(v_h, z) + P_c(v_z)}{\eta_p}, \qquad P_c(v_z) = \begin{cases} mgv_z/\eta_c, & v_z > \delta_z, \\ 0, & v_z \le \delta_z, \end{cases} \tag{21}$$

where $\delta_z = 0.01\ \mathrm{m\,s^{-1}}$ is the small vertical-rate deadband used in evaluation. The routine numerically floors v_h at $v_{\text{stall}}/2$; this floor is inactive for trajectories satisfying the stated stall and flight-path-angle bounds. With H_f denoting the fuel lower heating value,

$$F = \frac{1}{H_f} \int_0^T P_f(v_h(t), v_z(t), z(t))\, \mathrm{d}t, \qquad H_f = 43 \times 10^6\ \mathrm{J\,kg^{-1}}. \tag{22}$$

Thus F is reported in kilograms. Since P_f already includes division by η_p, the efficiency must not be applied a second time when converting energy to fuel mass.

For numerical integration, each sampled segment is subdivided at spatial fractions $0 = \lambda_0 < \cdots < \lambda_K = 1$. Constant tangential acceleration gives

$$v_{ij} = \sqrt{(1-\lambda_j)v_i^2 + \lambda_j v_{i+1}^2}, \qquad \Delta t_{ij} = \frac{2\ell_i(\lambda_{j+1} - \lambda_j)}{v_{ij} + v_{i,j+1}}. \tag{23}$$

At subsegment midpoint $\mathbf{q}_{ij}^m$, the horizontal and vertical velocities are calculated from the corresponding displacement divided by Δt_{ij}. The three numerical metrics are

$$\begin{aligned} T_{\text{num}} &= \sum_{i,j} \Delta t_{ij}, \\ C_{\text{num}} &= \sum_{i,j} P_d(\mathbf{q}_{ij}^m) \Delta t_{ij}, \\ F_{\text{num}} &= \frac{1}{H_f} \sum_{i,j} P_f(v_{h,ij}, v_{z,ij}, z_{ij}^m) \Delta t_{ij}. \end{aligned} \tag{24}$$

Spatial midpoint evaluation of the nonlinear field and power terms is a quadrature approximation. Search edges use the same field and power functions, with their edge time distributed uniformly over the subsegments.

This fuel model is intended for internally consistent comparisons. It neglects fuel-induced mass change, bank-dependent induced-drag corrections, transient engine behavior and an explicit tangential-acceleration power term. These assumptions limit claims about absolute fuel prediction.

2.4. Anchor-based normalized mission-oriented objective

Time, exposure and fuel have different units and numerical scales. A fixed reference is therefore constructed from three single-objective searches on the same coarse graph, with identical mission endpoints, environment, constraints and search-stage evaluation rules. Denote the resulting paths by Γ_T, Γ_C and Γ_F, corresponding to the weight vectors $(1,0,0)$, $(0,1,0)$ and $(0,0,1)$ in (T,C,F) order. These are discrete search anchors; they are not independently optimized final continuous trajectories.

Evaluating all three metrics on each anchor gives the payoff table

$$A = \begin{bmatrix} T^{\text{search}}(\Gamma_T) & C^{\text{search}}(\Gamma_T) & F^{\text{search}}(\Gamma_T) \\ T^{\text{search}}(\Gamma_C) & C^{\text{search}}(\Gamma_C) & F^{\text{search}}(\Gamma_C) \\ T^{\text{search}}(\Gamma_F) & C^{\text{search}}(\Gamma_F) & F^{\text{search}}(\Gamma_F) \end{bmatrix}. \tag{25}$$

For objective $k \in \{T,C,F\}$, let the lower reference be the corresponding anchor's diagonal entry, and let the upper reference be the column maximum:

$$b_k = A_{kk}, \qquad n_k = \max_{a\in\{T,C,F\}} A_{ak}, \qquad \Delta_k = n_k - b_k > 0. \tag{26}$$

The upper reference is a payoff-table nadir approximation, not the maximum over all feasible paths or an exact continuous Pareto nadir. Degenerate reference ranges must be rejected before normalized planning; they are not silently replaced by an arbitrary denominator.

For a metric vector $\mathbf{f} = (T,C,F)$, define

$$\widehat{T} = \frac{T - b_T}{\Delta_T}, \qquad \widehat{C} = \frac{C - b_C}{\Delta_C}, \qquad \widehat{F} = \frac{F - b_F}{\Delta_F}. \tag{27}$$

The same reference values are retained across weights, coarse and fine searches, and final-trajectory reporting within a fixed experiment. The normalized values are not clipped to $[0,1]$: a refined or retimed trajectory can fall outside the three-anchor reference ranges. This does not invalidate the normalization, but it precludes interpreting the reference values as exact bounds on final continuous performance.

Mission preference is expressed through

$$J = w_T\widehat{T} + w_C\widehat{C} + w_F\widehat{F}, \qquad w_T, w_C, w_F \geq 0, \quad w_T + w_C + w_F > 0. \tag{28}$$

The coefficients specify relative preferences and need not sum to one. Multiplication of all weights by the same positive constant preserves path ordering for a fixed model. During graph search, the additive contribution of edge e is

$$c_e = \frac{w_T \Delta T_e}{\Delta_T} + \frac{w_C \Delta C_e}{\Delta_C} + \frac{w_F \Delta F_e}{\Delta_F}. \tag{29}$$

The lower-reference offsets are applied once to the complete path:

$$J^{\text{search}}(\Gamma_d) = \sum_{e \in \Gamma_d} c_e - \left(\frac{w_T b_T}{\Delta_T} + \frac{w_C b_C}{\Delta_C} + \frac{w_F b_F}{\Delta_F} \right). \tag{30}$$

The bracketed term is path independent. Subtracting the lower references on every edge would instead introduce a spurious dependence on the number of edges and would not implement Eq. (28).

This scalarization directs discrete route selection. Spline candidates are subsequently ranked with a separate geometric refinement score, and the final (T, C, F, J) values are recomputed using the assigned speed profile. The sequential procedure is therefore not a joint global optimization of route geometry, speed and fuel, nor does it establish global Pareto optimality of the final trajectories.

The sensitivity experiment fixes $w_C = 1$ and evaluates $w_T \in \{0, 0.1, \ldots, 1.1\}$ and $w_F \in \{0, 0.1, \ldots, 0.5\}$, giving 72 configurations. The coefficients are swept rather than tuned by an additional optimizer. For tractability, this sweep uses coarse search followed by the same types of spline refinement, feasibility checks and final evaluation; it does not repeat the fine-corridor search for all 72 configurations. Consequently, it characterizes preference sensitivity of that coarse-search pipeline. Claims about the full two-stage pipeline require separate controlled evaluations that include the fine-corridor search.

3. Proposed risk-aware two-stage trajectory planning framework

The framework separates route selection, geometric refinement and speed assignment while retaining a common mission evaluator. The coarse search first identifies a route under the normalized time–exposure–fuel objective. A

fine search then resolves its local geometry within a three-dimensional corridor. Cubic B-splines replace the resulting polyline with differentiable candidates, which are screened against geometric constraints and ranked using a dimensionless refinement score. Finally, a forward–backward pass assigns speeds along the selected curve, and the mission metrics are recomputed under the resulting motion. This decomposition limits the size of the search state while making the distinction between search estimates and final trajectory performance explicit.

3.1. Multi-objective risk-aware A search*

The A* search [1] represents position on an anisotropic Cartesian lattice with horizontal and vertical spacings δ_{xy} and δ_z. Candidate displacements are drawn from

$$\mathcal{D} = \{(a\delta_{xy}, b\delta_{xy}, c\delta_z) : a, b, c \in \{-1, 0, 1\},\ (a, b, c) \neq (0, 0, 0)\}. \tag{31}$$

Purely vertical moves are excluded, and a displacement is retained only if its climb angle satisfies the prescribed bound. The map boundary, altitude envelope and hard exclusion regions further restrict the admissible graph. Each retained edge is checked at ten spatial locations, including its endpoints; the nine intervening subsegments supply the midpoint quadrature for exposure and fuel. A graph state contains position only. Heading, curvature and acceleration feasibility are consequently resolved after search rather than represented as additional state variables.

For a partial path ending at $\mathbf{p}$, the accumulated search cost is $G(\mathbf{p}) = \sum_e c_e$, with c_e defined in Eq. (29). The A* priority is

$$f(\mathbf{p}) = G(\mathbf{p}) + h(\mathbf{p}), \qquad h(\mathbf{p}) = \frac{w_T}{\Delta_T} \frac{\|\mathbf{p} - \mathbf{p}_g\|}{v_{\mathrm{cr}}}. \tag{32}$$

No heuristic inflation is used. The heuristic retains only a lower bound on the time contribution: risk and fuel are nonnegative, and the search time of an edge is at least its length divided by v_{cr}. For adjacent states $\mathbf{p}$ and $\mathbf{q}$, the triangle inequality therefore gives

$$h(\mathbf{p}) - h(\mathbf{q}) \leq \frac{w_T \|\mathbf{p} - \mathbf{q}\|}{\Delta_T v_{\mathrm{cr}}} \leq c_{(\mathbf{p},\mathbf{q})}. \tag{33}$$

Thus, the heuristic is consistent for the fixed graph and edge model. When $w_T = 0$, it becomes zero and the primary search reduces to uniform-cost expansion. Equal primary priorities are ordered using accumulated geometric

length plus Euclidean distance to the goal, providing a deterministic geometric preference in flat-cost regions. This secondary criterion does not add a distance penalty to the mission objective.

The constant lower-reference offset in Eq. (30) is restored when reporting the complete path objective. It is not included repeatedly in edge costs. The resulting minimum-cost interpretation concerns the discretized search problem, subject to numerical tolerances and successful search termination. It does not extend to the subsequently modified and retimed continuous trajectory.

3.2. Coarse-to-fine three-dimensional planning

A uniform fine lattice over the entire workspace requires many expansions in regions that are irrelevant to a particular mission preference. The two-stage search instead uses a coarse path Γ_c to restrict the domain of a second, finer search. For any polyline Γ, define

$$d(\mathbf{p}, \Gamma) = \min_{[\mathbf{a},\mathbf{b}]\in\Gamma} \min_{\tau\in[0,1]} \|\mathbf{p} - \mathbf{a} - \tau(\mathbf{b} - \mathbf{a})\|. \tag{34}$$

This distance uses the closest point on a segment, rather than the closest waypoint, and treats horizontal and vertical displacement in the same three-dimensional metric. The search corridor is

$$\mathcal{C}(\Gamma_c, r_c) = \{\mathbf{p} \in \mathcal{W} : d(\mathbf{p}, \Gamma_c) \le r_c\}. \tag{35}$$

Where grid snapping changes an endpoint, the physical endpoint is connected to the coarse polyline when constructing the corridor reference. The endpoints used in the present experiments lie on both lattices.

The fine graph retains nodes inside this corridor and samples candidate edges to prevent a connection between admissible endpoints from leaving the corridor unchecked. For the reported settings, corridor probes are spaced no more than approximately 100 m along an edge, with at least three probes. Altitude and obstacle screening remain active. Both search stages use the same objective weights and the same anchor-derived normalization scales; the fine stage does not recompute its own anchors.

The fine result Γ_f is obtained by a new graph search, not by interpolating Γ_c. Its finer directional and altitude resolution permits local route changes, but the corridor limits which alternatives can be explored. The construction therefore trades search coverage for computational tractability; it does not guarantee equivalence to an unrestricted fine-grid search. A failed fine search is reported as a failure rather than relabeling the coarse path as a fine result.

3.3. Cubic B-spline trajectory refinement

The fine polyline satisfies the edge-level search constraints but retains direction discontinuities at its vertices. Let $\{\mathbf{P}_i\}_{i=0}^{n-1}$ denote its ordered vertices after removal of consecutive duplicates. A cubic parametric B-spline is fitted as

$$\mathbf{q}_\sigma(u) = \sum_{j=0}^{K_\sigma - 1} B_{j,3}(u)\mathbf{c}_j, \qquad u \in [0,1], \tag{36}$$

where $B_{j,3}$ are cubic basis functions, $\mathbf{c}_j$ are control points, and σ controls the fit. Chord-length parameterization assigns $u_i = \sum_{k=0}^{i-1} \|\mathbf{P}_{k+1} - \mathbf{P}_k\| / L_{\text{raw}}$, where L_{raw} is the polyline length. The initial FITPACK smoothing fit [6] uses the weighted residual condition

$$\sum_{i=0}^{n-1} \omega_i^2 \|\mathbf{P}_i - \mathbf{q}_\sigma(u_i)\|^2 \le \sigma, \qquad \omega_i = \begin{cases} 10^5, & i \in \{0, n-1\}, \\ 1, & \text{otherwise.} \end{cases} \tag{37}$$

Large endpoint weights promote endpoint retention, which is subsequently verified with an explicit tolerance. The smoothing condition controls the initial fit; it is not the final mission objective and is not assumed to remain an exact residual bound after geometric repair.

To explore the balance between polyline fidelity and curvature reduction, the smoothing budgets are scaled to the input resolution:

$$\sigma_j = n(\delta_{xy}\beta_j)^2. \tag{38}$$

The fixed set of 21 dimensionless factors is

$$\begin{aligned} \mathcal{B} = \{&0.02, 0.03, 0.04, 0.05, 0.0625, 0.075, 0.09, \\ &0.105, 0.125, 0.15, 0.18, 0.22, 0.25, 0.35, \\ &0.50, 0.75, 1.0, 1.5, 2.0, 3.0, 4.0\}. \end{aligned} \tag{39}$$

Every budget is examined before selecting a candidate. Small budgets retain the lattice geometry more closely; larger budgets allow greater deformation but must still satisfy the local-deviation and flight-envelope tests. The input is Γ_f for the complete two-stage method and Γ_c for the separate coarse-search sensitivity protocol.

Geometric repair addresses constraint violations introduced by smoothing. Altitude repair raises deficient sampled altitudes relative to the spatial altitude floor and refits the corrected points. Climb repair limits the vertical

slope using margins below $\tan\gamma_{\max}$ while retaining the sampled horizontal coordinates before refitting. These operations generate revised candidates, each of which is rechecked against the full geometric test set. Repair is a feasibility procedure and need not reduce the mission objective.

The terminal heading is inferred from the incoming raw route over a lookback length of $\max(5\delta_{xy}, 1000\text{ m})$, using the horizontal chord from the selected upstream vertex to the goal. If the initial candidate pass finds no geometrically feasible curve and includes a heading violation, a second pass aligns the final control-polygon edge with this reference and reevaluates all smoothing budgets. The adjustment retains the endpoint coordinates and control-point altitudes. The method thus preserves the approach direction implied by the route; it does not enforce an independently specified terminal mission heading.

For regular curves with simple interior knots, cubic B-splines provide C^2 geometric continuity and permit curvature evaluation from analytic derivatives. This property removes polyline corners, but does not by itself establish feasibility or imply continuous jerk in the time-parameterized motion. Those issues require the subsequent geometric and speed checks.

3.4. Geometric feasibility validation and candidate selection

Candidate curves are evaluated at $M = 4096$ uniformly spaced parameter values. Their analytic derivatives give spatial curvature through Eq. (12) and the local flight-path angle

$$\gamma(u) = \arctan\left(\frac{q_z'(u)}{\sqrt{q_x'(u)^2 + q_y'(u)^2}}\right). \tag{40}$$

Finite coordinates and nonzero tangents are required. The geometric turning-radius threshold is

$$R_{\text{ref}} = \max\left\{R_{\text{floor}}, \frac{v_{\text{stall}}^2}{g\tan\phi_{\max}}\right\}. \tag{41}$$

Requiring $\kappa \le 1/R_{\text{ref}}$ prevents acceptance of a turn that already exceeds the bank envelope at stall speed. It does not require the curve to support cruise speed everywhere, because speed can subsequently decrease in curved sections.

A candidate must also satisfy the altitude envelope, avoid the hard exclusion regions, retain both endpoints within 2 m, and match the terminal

horizontal heading within 12°. Its maximum distance from the input polyline must satisfy

$$d(\mathbf{q}_\sigma(u), \Gamma_{\text{raw}}) \le r_s \tag{42}$$

at the validation samples. Here r_s is a refinement locality limit, distinct from the coarse-search corridor radius r_c. In the default configuration, membership of the final spline in the coarse corridor is recorded diagnostically, whereas locality relative to the raw input curve is enforced. Obstacles and the altitude envelope remain hard requirements.

Among the geometrically admissible candidates, selection balances bending, curvature variation, deviation and length. Let s denote arc length, L_σ the candidate length, and $\kappa_s = \mathrm{d}\kappa/\mathrm{d}s$. The dimensionless components are

$$Q_b = \frac{1}{L_\sigma} \int_0^{L_\sigma} (R_{\text{ref}}\kappa)^2 \, \mathrm{d}s, \tag{43}$$

$$Q_v = R_{\text{ref}}^3 \int_0^{L_\sigma} \kappa_s^2 \, \mathrm{d}s, \tag{44}$$

$$Q_d = \frac{1}{r_s} \sqrt{\frac{1}{M} \sum_{m=0}^{M-1} d(\mathbf{q}_\sigma(u_m), \Gamma_{\text{raw}})^2}, \qquad u_m = \frac{m}{M-1}, \tag{45}$$

$$Q_l = \frac{L_\sigma}{L_{\text{raw}}}. \tag{46}$$

The integrals use the cumulative chord-length coordinate of the dense samples and trapezoidal quadrature; κ_s is estimated numerically on that coordinate. In contrast, Q_d is the root-mean-square deviation over uniform parameter samples, not an arc-length-weighted integral. The curvature-variation term penalizes spatial changes in curvature; it is not a roll-rate constraint.

The refinement score and selection rule are

$$J_s = 0.4Q_b + 0.3Q_v + 0.2Q_d + 0.1Q_l, \qquad \mathbf{q}^* = \operatorname*{arg\,min}_{\mathbf{q} \in \mathcal{Q}_{\text{geom}}} J_s(\mathbf{q}), \tag{47}$$

where $\mathcal{Q}_{\text{geom}}$ contains the admitted candidates from the applicable fitting and repair pass. The four coefficients are fixed geometric design settings, independent of (w_T, w_C, w_F). Selection enumerates the candidate set rather than stopping at the first feasible smoothing budget. It is a finite candidate comparison, not a global optimization over all spline control points.

If no geometrically admissible candidate exists, refinement fails. A diagnostic fallback curve is not accepted as a mission trajectory. The selected

curve must next pass strict speed validation; the present procedure does not rerank alternative candidates following a speed-stage failure. Final acceptance therefore requires both stages to succeed. Because geometric tests use finite sampling, acceptance has the numerical meaning stated in Section 2.2.

3.5. Adaptive speed planning and final mission evaluation

The selected spline is resampled at approximately uniform arc-length spacing. A dense set of 8000 parameter probes supplies the cumulative length map, which is inverted to obtain the final samples $\{\mathbf{q}_i\}_{i=0}^{N}$. Their chord lengths are $\ell_i = \|\mathbf{q}_{i+1} - \mathbf{q}_i\|$, and κ_i is evaluated from the spline derivatives at the corresponding parameters. The target spacing is 100 m in both experimental protocols.

The vertical-rate ceiling of a nondegenerate segment is

$$\bar{v}_{z,i} = \begin{cases} v_{\mathrm{asc,max}}\ell_i/|z_{i+1} - z_i|, & z_{i+1} > z_i, \\ v_{\mathrm{des,max}}\ell_i/|z_{i+1} - z_i|, & z_{i+1} < z_i, \\ +\infty, & z_{i+1} = z_i. \end{cases} \tag{48}$$

This ceiling is assigned to both endpoints of the segment. At node i, the admissible speed ceiling is

$$\bar{v}_i = \min\left\{ v_{\mathrm{cr}}, v_{\max}, \sqrt{\frac{g\tan\phi_{\max}}{\kappa_i}}, \min_{e\in\mathcal{E}_i} \bar{v}_{z,e} \right\}, \tag{49}$$

where $\mathcal{E}_i$ comprises adjacent segments and the curvature term is infinite on a straight segment. A ceiling below v_{stall} causes rejection, rather than a speed assignment below stall.

With $v_0^+ = v_{\mathrm{stall}}$, the forward pass propagates the acceleration bound:

$$v_{i+1}^+ = \min\left\{ \bar{v}_{i+1}, \sqrt{(v_i^+)^2 + 2a_{\mathrm{acc,max}}\ell_i} \right\}, \qquad i = 0, \ldots, N-1. \tag{50}$$

The backward pass anticipates downstream speed reductions:

$$v_N = v_N^+, \qquad v_i = \min\left\{ v_i^+, \sqrt{v_{i+1}^2 + 2a_{\mathrm{dec,max}}\ell_i} \right\}, \qquad i = N-1, \ldots, 0. \tag{51}$$

No terminal-speed equality is introduced. Applying a segment's vertical ceiling at both ends bounds its largest vertical speed under the adopted

constant-tangential-acceleration segment model. The final validation checks speed, turning, climb angle, vertical rate and tangential acceleration; failure prevents the trajectory from being reported as feasible.

The accepted speed sequence supplies segment times through Eq. (16). Time, exposure and fuel are then evaluated with the common subsegment quadrature in Eq. (24), including renewed altitude and collision screening, and normalized using the unchanged anchor scales. This last evaluation is essential because spline deformation changes the spatial threat field sampled by the vehicle, while retiming changes both residence time and fuel expenditure. Consequently, a lower geometric refinement score does not by itself imply a lower final mission objective. The reported output is the accepted geometry, its assigned speed profile and the recomputed (T, C, F, J); search-stage estimates are retained separately for diagnosis and stage comparisons.

4. Numerical results and discussion

The evaluation first separates the effects of mission-cost modeling, spatial refinement and trajectory generation through controlled comparisons within the same planning framework. A 72-case experiment then examines mission-preference sensitivity, from which three representative preferences are selected to illustrate the observed time–exposure–fuel trade-offs. Search-stage estimates and final trajectory metrics are distinguished throughout the analysis.

4.1. Simulation setup and evaluation protocols

The simulation uses a 100×100 km horizontal workspace containing five ground-based radar sites and five associated cylindrical exclusion zones. The radar layout is generated with seed 42 and held fixed throughout the preference comparison. Each radar has a hemispherical coverage radius of 15 km, while its colocated hard zone has a horizontal radius of 6 km and a height of 5 km. The power-law exposure model uses an inner-radius ratio of 0.3 and an exponent of 4, with overlapping fields combined by their maximum. These settings define a controlled synthetic environment; they are not calibrated measurements of a particular radar or aircraft system.

The mission endpoints are $(2, 2, 0)$ and $(98, 98, 0)$ km. The en-route altitude interval is 1–10 km, with a 5 km horizontal transition distance at each endpoint. The zero endpoint altitude is the reference used for the airborne

Table 1: Parameters of the reduced fixed-wing model.

Parameter	Value
Constant mass	1100 kg
Wing reference area	11.5 m^2
Gravitational acceleration	9.81 $\mathrm{m\,s}^{-2}$
Stall / cruise / aircraft maximum speed	33/55/83 $\mathrm{m\,s}^{-1}$
Maximum ascent / descent rate	12/15 $\mathrm{m\,s}^{-1}$
Tangential acceleration / deceleration bound	3/4 $\mathrm{m\,s}^{-2}$
Maximum bank / flight-path angle	30°/15°
Structural turning-radius floor	50 m
Zero-lift drag coefficient / aspect ratio	0.025/18
Oswald efficiency	0.85
Climb / overall propulsive efficiency	0.80/0.30
Descent energy recovery fraction	0
Fuel lower heating value	43 $\mathrm{MJ\,kg}^{-1}$
Sea-level density / density scale height	1.225 $\mathrm{kg\,m}^{-3}$/8500 m

trajectory model, as described in Section 2.2. The vehicle and energy parameters are listed in Table 1; the numerical planning settings are summarized in Table 2.

Two protocols address different experimental questions. The complete two-stage protocol comprises coarse search, corridor-restricted fine search, spline refinement and strict speed validation. It supports the controlled comparisons in Section 4.2. The sensitivity protocol uses coarse search followed by refinement and speed validation for each of the 72 weight configurations specified in Section 2.4. These results support both the sensitivity analysis and the representative-preference comparison in Sections 4.3 and 4.4. Its locality limit is scaled to the coarse cell diagonal,

$$r_s^{\mathrm{sweep}} = 1.05\sqrt{2(2000)^2 + (500)^2} \simeq 3015.90 \text{ m}, \tag{52}$$

whereas the complete method uses $r_s = 200$ m relative to the fine polyline. The two protocols share the threat and aircraft models, objective scales, candidate-ranking rule and final metric definitions. Their different search domains and locality limits are retained explicitly when interpreting the results.

Search-stage and final-trajectory quantities are distinguished throughout the analysis. Flight time T, exposure C and fuel F are reported in seconds, exposure-seconds and kilograms, respectively; path length is reported

Table 2: Search and trajectory-evaluation settings. The sweep and the complete two-stage method use different input resolutions and locality limits.

Parameter	Value
Coarse horizontal / vertical spacing	2000/500 m
Fine horizontal / vertical spacing	200/50 m
Three-dimensional fine-search corridor radius	600 m
Spline locality limit, fine input	200 m
Spline locality limit, coarse sweep input	3015.90 m
Search edge probes / quadrature subsegments	10/9
A* heuristic inflation factor	1.0
Initial smoothing budgets	21
Spline geometric validation samples	4096
Arc-length inversion probes	8000
Final target sampling distance	100 m
Endpoint / terminal-heading tolerance	2 m/12°
Initial speed	33 $\mathrm{m\,s^{-1}}$
Terminal speed	No prescribed equality

in kilometres. The exposure integral measures time-weighted presence in the prescribed field, not a mission-level detection probability. Cross-preference comparisons use the physical vector (T, C, F): a value of J computed with one weight vector cannot directly rank a trajectory scored with another. For a controlled comparison at fixed weights, the shared anchor normalization provides the common scalar objective.

4.2. Overall planning performance and controlled comparisons

Three comparisons address distinct parts of the planning chain. The first examines the effect of the mission objective, the second compares coarse and fine raw routes from one run, and the third assesses the transition from that fine route to a continuous trajectory with an assigned speed profile. The latter two comparisons reuse intermediate outputs rather than requiring separate planning experiments. These are controlled objective and stage comparisons, rather than a complete factorial ablation: each comparison is interpreted at the level of the factors it actually changes.

4.2.1. Time-oriented A versus proposed planning*

The reference is a time-oriented A* planner with weights $(1, 0, 0)$. Hard-zone admissibility, altitude and climb constraints are retained, whereas con-

tinuous soft exposure and fuel do not enter its search cost. The proposed formulation uses the normalized composite cost in Eq. (28). Thus, the comparison concerns the information encoded in the objective, rather than a limitation of A* to binary costs. The radar exposure field remains traversable for both formulations and is evaluated using the same model after planning.

Both formulations use the same environment, aircraft parameters, lattice resolutions and refinement and speed-assignment procedures. Their coarse raw paths can first be compared on the common coarse graph. Final mission performance is then compared after each route has completed the two-stage pipeline and passed its feasibility checks. Each route defines its own fine corridor, so this final comparison measures the end-to-end effect of the objective rather than holding the fine search domain fixed.

The principal measures are path length L, flight time T, accumulated exposure C and fuel consumption F. The analysis concerns their trade-off, not simultaneous improvement in all four quantities. A lower exposure accompanied by additional time or fuel would indicate the cost of the selected mission preference, if observed in the data. A scalar comparison of J requires both trajectories to be evaluated under one stated weight vector and the same anchor references.

4.2.2. Coarse versus fine planning

The coarse raw route and the fine raw route are extracted from the same main run. Both use identical mission weights and anchor scales. Their comparison therefore concerns the local geometric changes produced by fine-grid replanning within the three-dimensional corridor, before any spline deformation or speed reassignment. Spatial overlays and altitude profiles identify where the fine route departs from the coarse reference.

Search-stage L, T, C, F and J are reported together with the stage-specific computation times and expansion counts. These observations assess the practical effect and cost of local refinement; improvement in every metric is not presumed. Without an unrestricted fine-grid reference, the results do not establish a quantified speedup over full-domain fine search or a globally optimal route topology.

4.2.3. Fine raw path versus B-spline trajectory

The fine raw path and its refined trajectory are obtained from the same run. This comparison addresses the distinction between an admissible piecewise-linear route and a continuous trajectory that passes the adopted fixed-wing

flight-envelope checks. Local geometry, endpoint and heading retention, altitude and hard-zone screening are examined together with minimum turning radius, analytic spline curvature and flight-path angle. Curvature inferred from three raw vertices is a discretization diagnostic, not continuous curvature at a polyline corner.

The final speed $v(s)$ and signed tangential acceleration $a(s)$ are assessed against the stall, local curvature, vertical-rate and acceleration limits. The initial speed is prescribed, while no terminal-speed equality is imposed. This evaluation concerns numerical compliance with the reduced flight-envelope model, not validation of a complete aircraft control system. Final T, C and F are recomputed along the refined geometry and its assigned speed profile. Their simultaneous reduction is not a requirement: smoothing and retiming serve feasibility, and the geometric selection score J_s is distinct from the mission objective J.

The relation between geometry and exposure is examined along the final trajectory. Since $dt = ds/v(s)$, a section with elevated threat intensity contributes more exposure when its traversal speed is lower. Curvature- or vertical-rate-induced speed reductions can therefore modify the mission trade-off after route selection. Aligning the threat intensity, curvature and speed profiles by arc length helps identify whether such effects occur in the selected case. This interpretation concerns the observed trajectory; it does not attribute the entire raw-to-final cost change to a single processing stage.

4.3. Mission-preference sensitivity

The preference experiment fixes $w_C = 1$ and varies

$$w_T \in \{0, 0.1, \ldots, 1.1\}, \qquad w_F \in \{0, 0.1, \ldots, 0.5\}, \tag{53}$$

resulting in $12 \times 6 = 72$ configurations. Each configuration follows the coarse-search, spline-refinement and speed-assignment protocol in Section 4.1. The same environment, aircraft model, anchor references and final evaluation rules are retained. In particular, $(w_T, w_F) = (0, 0)$ is an exposure-only objective because w_C remains one.

Three heatmaps summarize the final time, exposure and fuel responses to the two varied coefficients. A fourth plot presents the T–C distribution with fuel consumption encoded by color. Together, these views describe the objective trade-offs without comparing scalar scores computed under different preferences. Plateaus and abrupt changes, where present, should be examined in relation to route selection and subsequent refinement; monotonic

changes in the final metrics are not assumed. The experiment characterizes the sampled preferences in the stated environment, rather than a global Pareto frontier or performance across independent scenarios.

4.4. Representative mission preferences

Three selection rules identify risk-dominant, time-dominant and balanced preferences from the same 72-case dataset. For case index $i \in \mathcal{I}$, the first two rules select

$$i_R \in \arg\min_{i\in\mathcal{I}} C_i, \qquad i_T \in \arg\min_{i\in\mathcal{I}} T_i. \tag{54}$$

For the balanced rule, each physical objective f_{ik}, $k \in \{T, C, F\}$, is scaled within the evaluated batch:

$$f'_{ik} = \begin{cases} \dfrac{f_{ik} - \min_{j\in\mathcal{I}} f_{jk}}{\max_{j\in\mathcal{I}} f_{jk} - \min_{j\in\mathcal{I}} f_{jk}}, & \max_j f_{jk} > \min_j f_{jk}, \\ 0, & \text{otherwise.} \end{cases} \tag{55}$$

The balanced case minimizes the Euclidean distance to the batch ideal point,

$$D_i = \sqrt{(T'_i)^2 + (C'_i)^2 + (F'_i)^2}, \qquad i_B \in \arg\min_{i\in\mathcal{I}} D_i. \tag{56}$$

This batch scaling is used only for representative selection and does not replace the three-anchor normalization in the planner. Equal primary values are resolved lexicographically by (T, C, F, w_T, w_F) for the balanced rule, (T, F, w_T, w_F) for the exposure rule, and (C, F, w_T, w_F) for the time rule. These definitions match the implemented selection procedure.

The selected final trajectories are taken directly from the corresponding sweep caches. One three-dimensional overlay and one table summarize their geometry, weights and final (T, C, F) values under that common protocol. The labels describe selection rules within the sampled dataset, not independent operating environments or globally optimal mission solutions. If two rules select the same configuration, their coincidence is reported rather than forcing distinct cases. The balanced choice expresses equal importance in batch-scaled objective space; it is neither the minimizer of J across different weights nor a universally preferred mission setting.

5. Conclusions

This paper develops a risk-aware planning framework that connects mission-dependent graph search with fixed-wing trajectory generation. Hard no-fly

Table 3: Representative preferences selected from the 72-case coarse-search experiment. Metrics refer to the final refined and retimed trajectories.

Case	w_T	w_C	w_F	T (s)	C (exposure-s)	F (kg)
Risk-dominant	–	–	–	–	–	–
Time-dominant	–	–	–	–	–	–
Balanced	–	–	–	–	–	–

constraints and continuous soft exposure have distinct roles in the model, while time, exposure and fuel are combined through shared reference scales. Coarse and corridor-constrained fine searches retain the same mission objective. Subsequent spline refinement and speed assignment convert the selected path into a trajectory evaluated against the prescribed flight envelope. Reintegrating the mission metrics after these stages preserves the distinction between search estimates and the cost of the resulting motion.

The scope of the study is a prescribed static environment and a reduced aircraft model. The exposure integral is a planning surrogate, and the fuel estimate is evaluated under the stated aerodynamic and efficiency assumptions. Numerical feasibility checks do not constitute a continuous collision certificate or a closed-loop flight demonstration. Within this scope, the framework provides an auditable basis for relating mission preferences to the geometry, motion limits and evaluated cost of a planned trajectory. Further work will examine the effect of environmental and model uncertainty and connect the generated references with higher-fidelity flight dynamics and trajectory tracking.

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